

The above observed decreasing trends of compressive properties with increasing mesostructural stochastics can be explained through the weakest link principle (see also [61, 62]). Introducing the stochastic variations of more mesostructural features gives rise to the emergence of more weak regions. Indeed, Figure 19 demonstrates that the cell wall buckling events (using eq. (36)) tend to get promoted as more mesostructural stochastics are included (from “CtCt” to “StSt”). Here, N_c and N_w denote the number of buckled cell walls and the total number of cell walls, respectively. The fractions N_c/N_w of buckled cell walls for different model sets eventually become comparable upon a large applied strain. As expected, the cell wall buckling events for H200 are postponed (see also Figure 17) and accompanied by a slower growth of N_c/N_w , compared with H100.

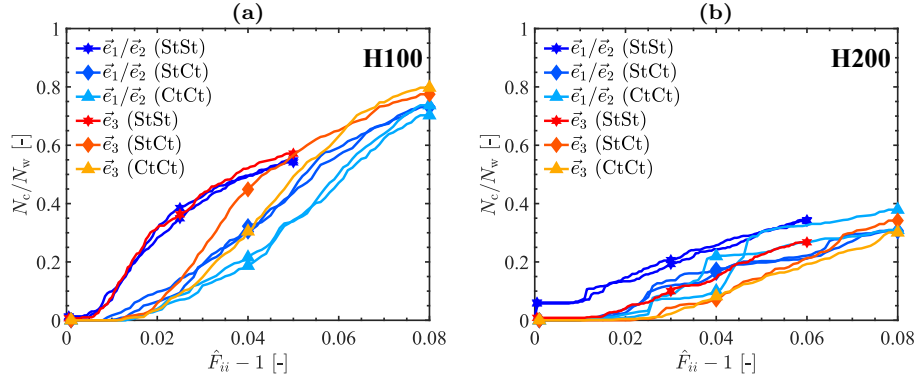


Figure 19: Fractions of buckled cell walls versus applied strain of different tessellation-based model sets for two Divinycell foam grades, under uniaxial compression in the transverse ($\vec{e}_1/\vec{e}_2$) and foam rise ($\vec{e}_3$) directions.

For the sake of reference, the experimental compressive properties of H100 [97] and H200 [98], are provided in Figure 18 (leftmost bars). $\hat{\nu}_{13}^*$ are not measured and $\hat{\nu}_{31} = \hat{\nu}_{32}$ has been assumed in [97, 98]. As an indication, the experimental data from other literature are also provided in Figure 18 (black crosses), although these studies are lacking either well-defined strain measurements or complete stress-strain curves under uniaxial compression. A remarkable inconsistency between the experimental data reported in different literature can be noticed. This places a clear need of more attention to the experimental aspects, e.g. test method, sample shape, sample size and determination of compressive properties.

In the following, the numerical model predictions are compared with the experimental data from [97, 98] only, given their reliability and relevance. The model set “StSt”, with all the cell size, cell wall thickness and cell shape anisotropy stochastics incorporated, seems to deliver the closest predictions with respect to the experimental data. In particular for the compressive moduli and Poisson’s ratios, an excellent agreement between the experimental data and “StSt” predictions can be observed. Relatively large deviations appear on the strengths, which are overestimated by $\sim 15\%$, likely due to the disregarded plasticity in the present numerical models.

7.3. Mechanical anisotropy

With the effective compressive properties in Figure 18, the mechanical anisotropy $\mathcal{R}^E$ and $\mathcal{R}^\sigma$ of different model sets are computed, and reported in Figures 20(a) and (b), respectively. Three model sets (“StSt”, “StCt” and “CtCt”), are found to deliver comparable predictions of both $\mathcal{R}^E$ and $\mathcal{R}^\sigma$, with the relative difference in between $< 10\%$. This suggests that the cell wall thickness and cell size stochastics only weakly affect the resulting mechanical anisotropy.